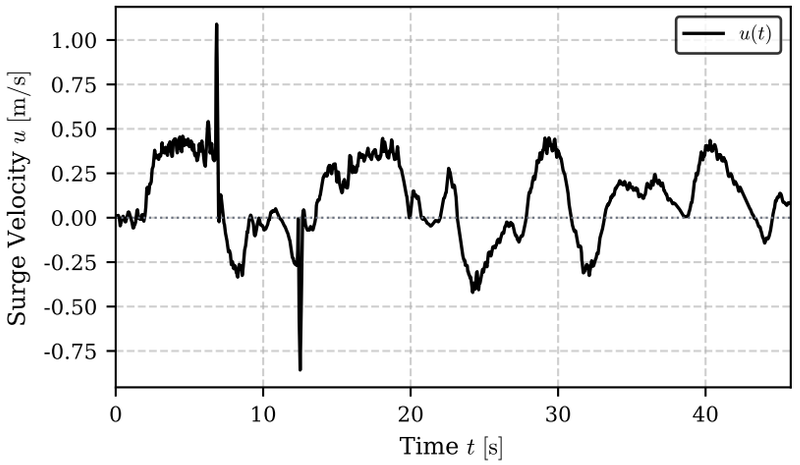
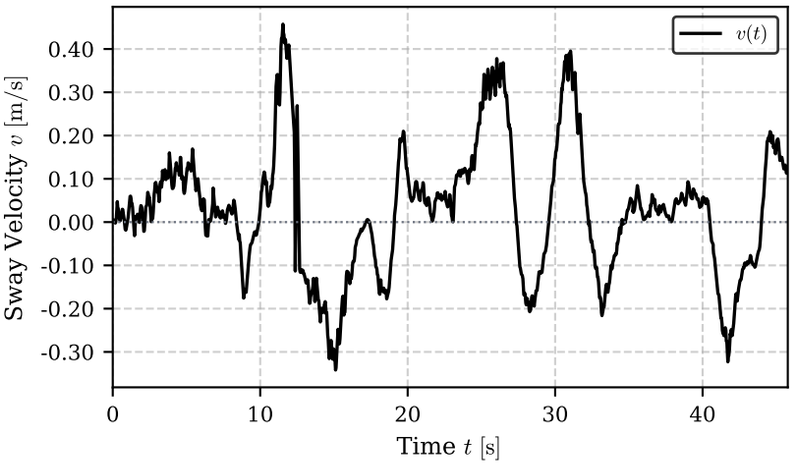
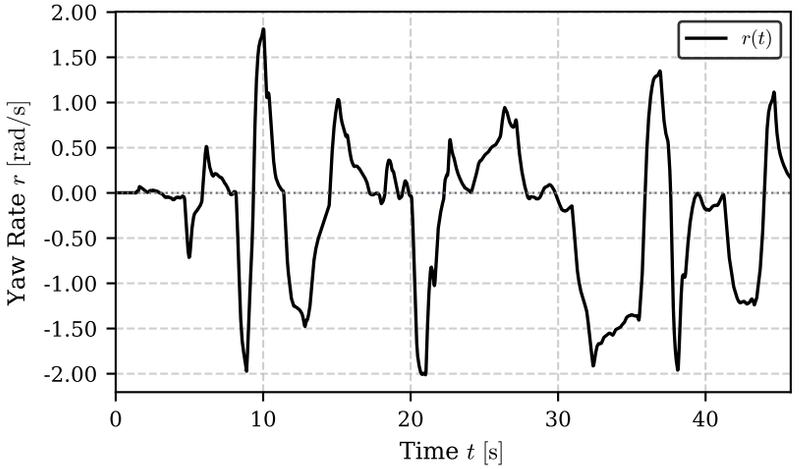
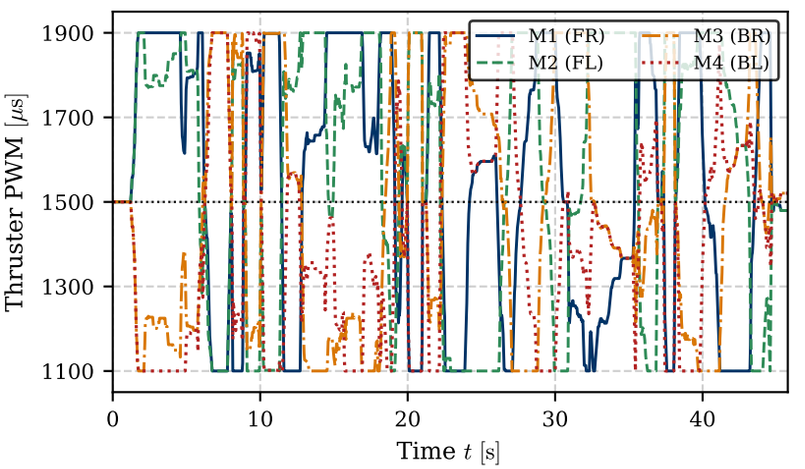
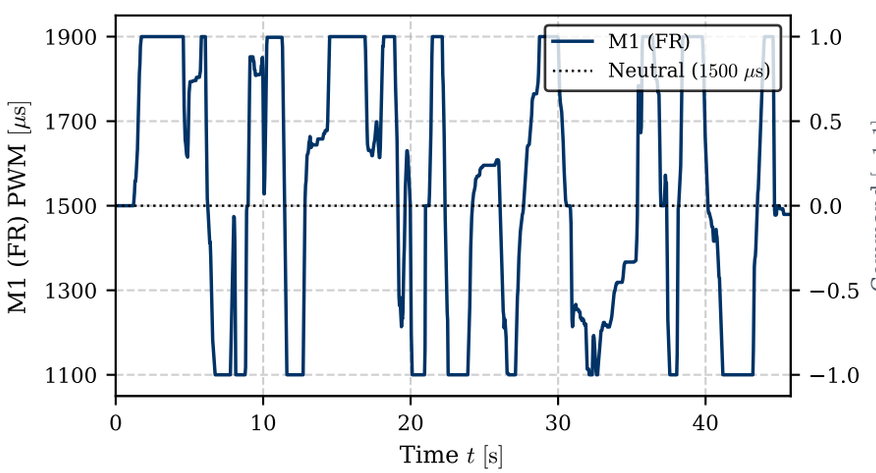
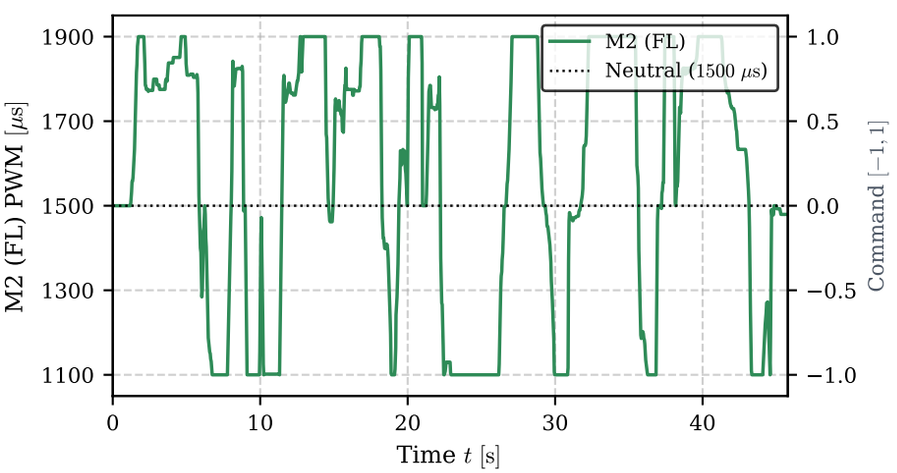

$u(t)$
Surge Velocity u [m/s]
Time t [s]

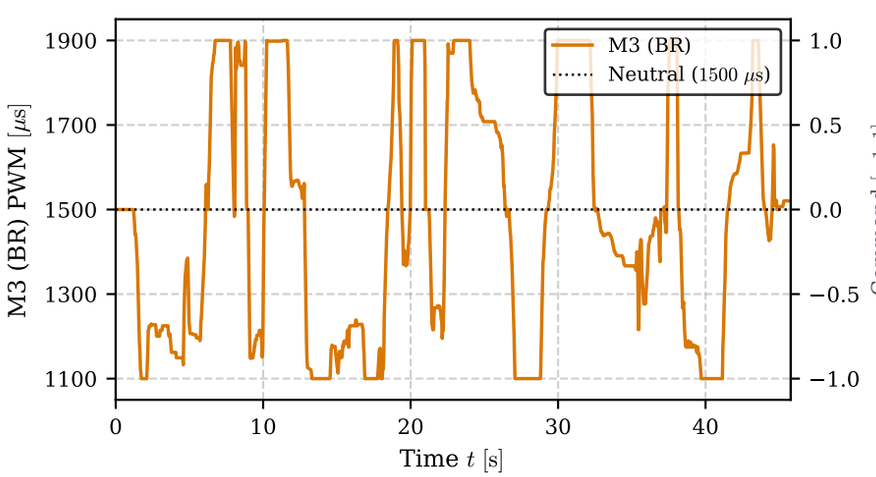

$v(t)$
Sway Velocity v [m/s]
Time t [s]

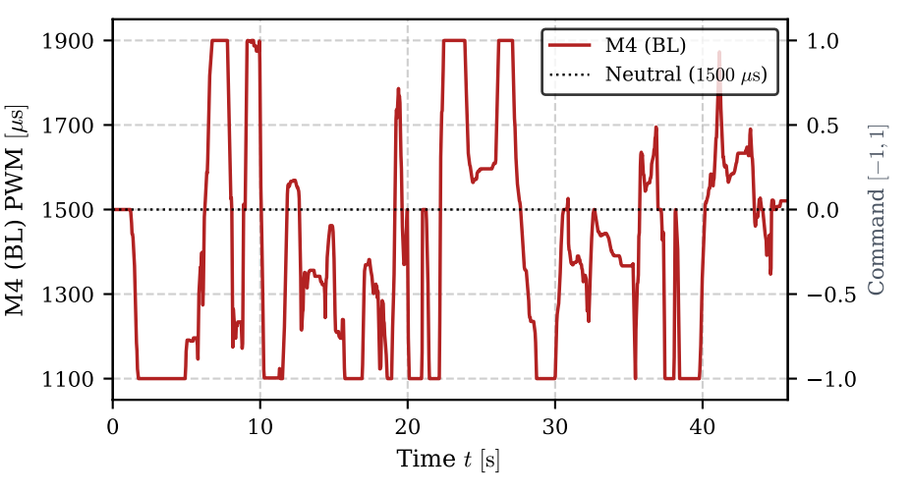

$r(t)$
Yaw Rate r [rad/s]
Time t [s]

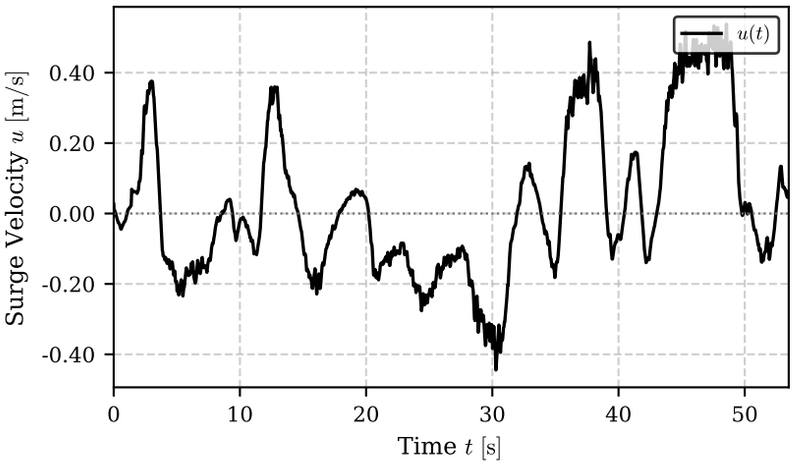

M1 (FR)
M2 (FL)
M3 (BR)
M4 (BL)
Thruster PWM [μs]
Time t [s]

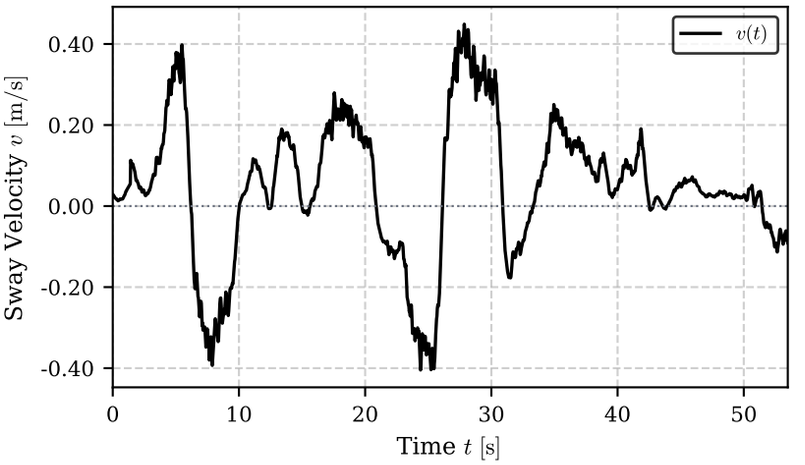

M1 (FR)
Neutral (1500 μs)
M1 (FR) PWM [μs]
Command $[-1, 1]$
Time t [s]

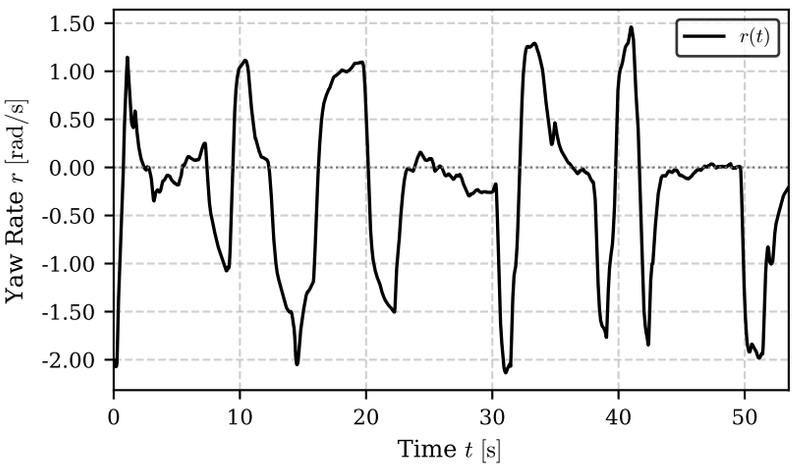

M2 (FL)
Neutral (1500 μs)
M2 (FL) PWM [μs]
Command $[-1, 1]$
Time t [s]

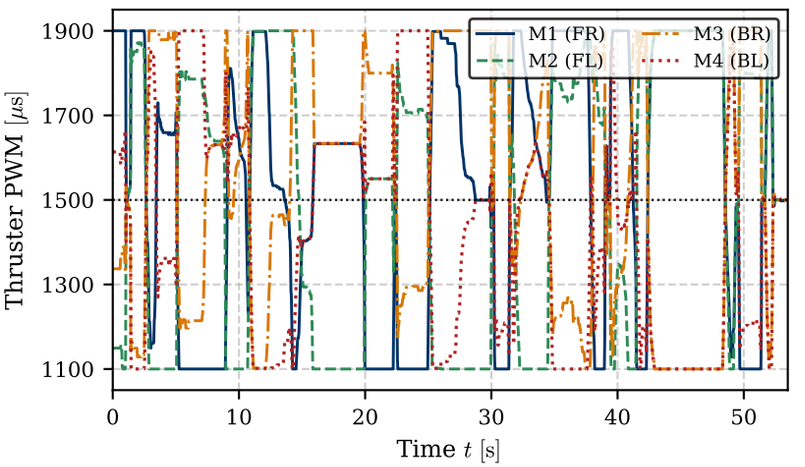

M3 (BR)
Neutral (1500 μs)
M3 (BR) PWM [μs]
Command $[-1, 1]$
Time t [s]

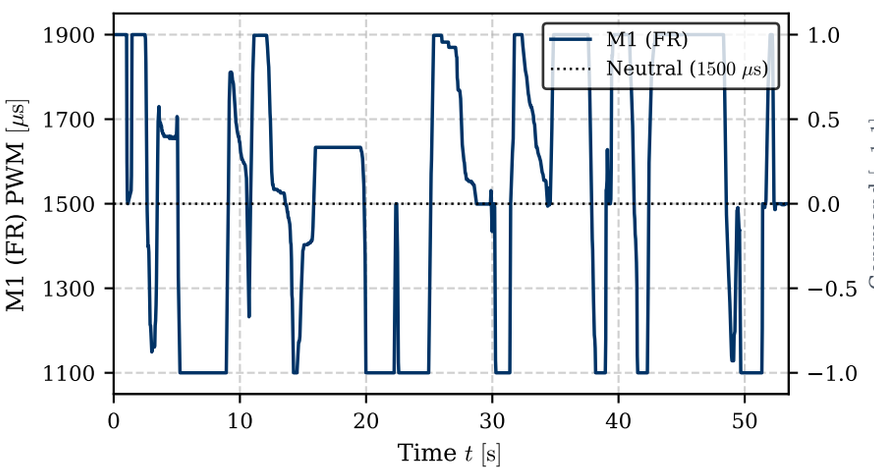

M4 (BL)
Neutral (1500 μs)
M4 (BL) PWM [μs]
Command $[-1, 1]$
Time t [s]

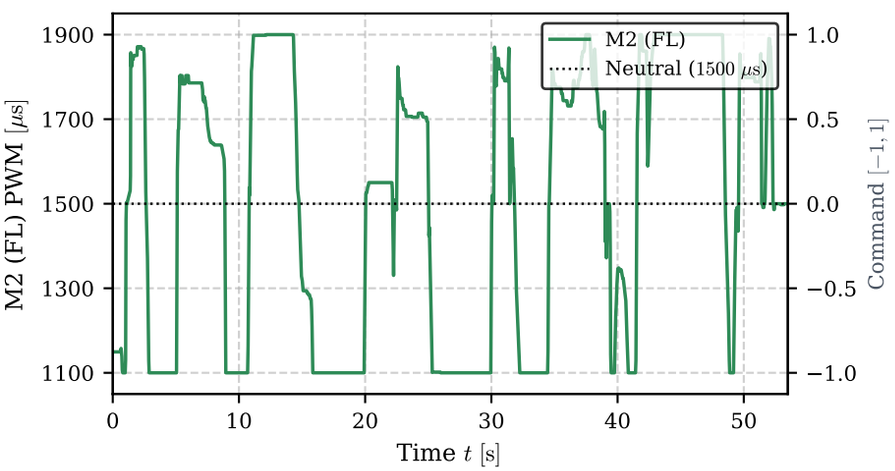

$u(t)$
Surge Velocity u [m/s]
Time t [s]

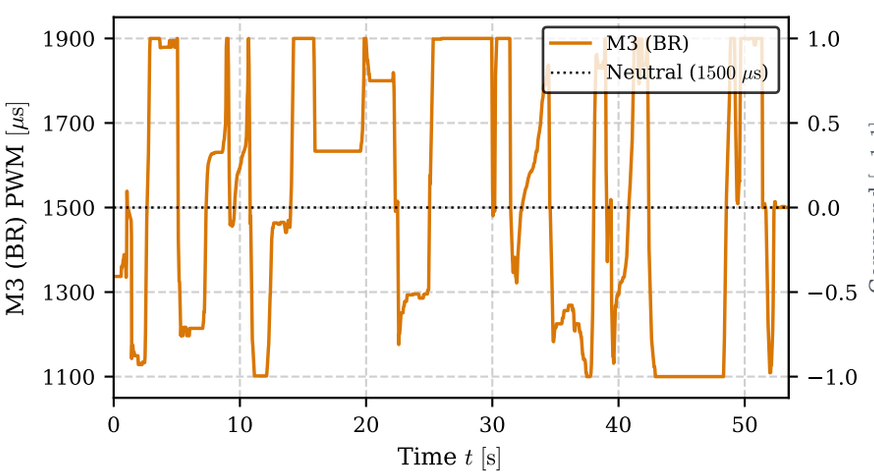

$v(t)$
Sway Velocity v [m/s]
Time t [s]

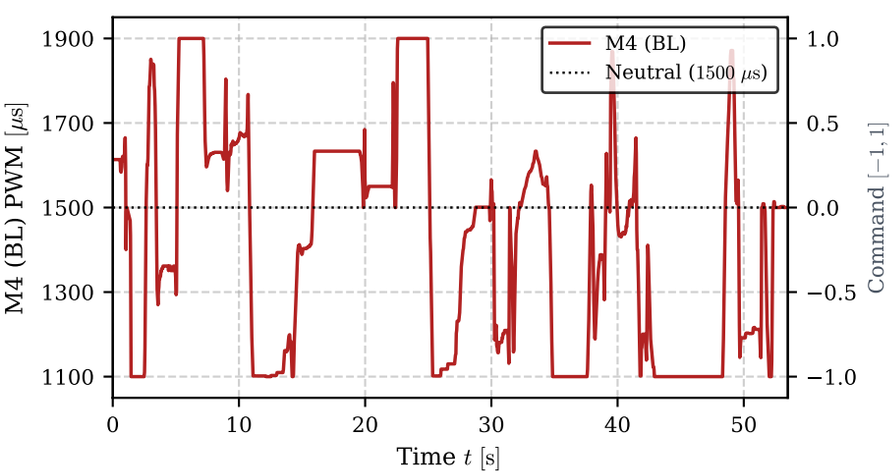

$r(t)$
Yaw Rate r [rad/s]
Time t [s]

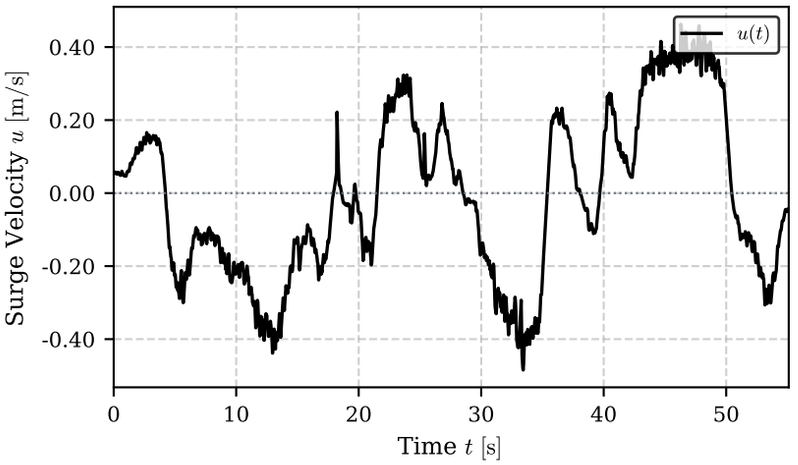

M1 (FR)
M2 (FL)
M3 (BR)
M4 (BL)
Thruster PWM [μs]
Time t [s]

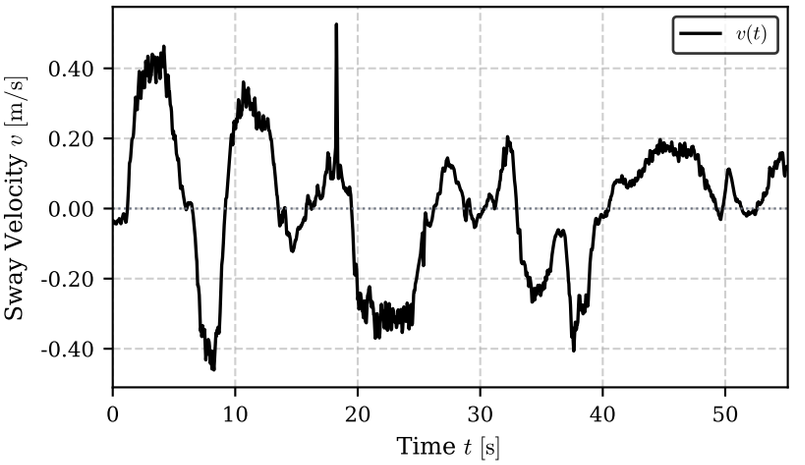

M1 (FR)
Neutral (1500 μs)
M1 (FR) PWM [μs]
Command $[-1, 1]$
Time t [s]

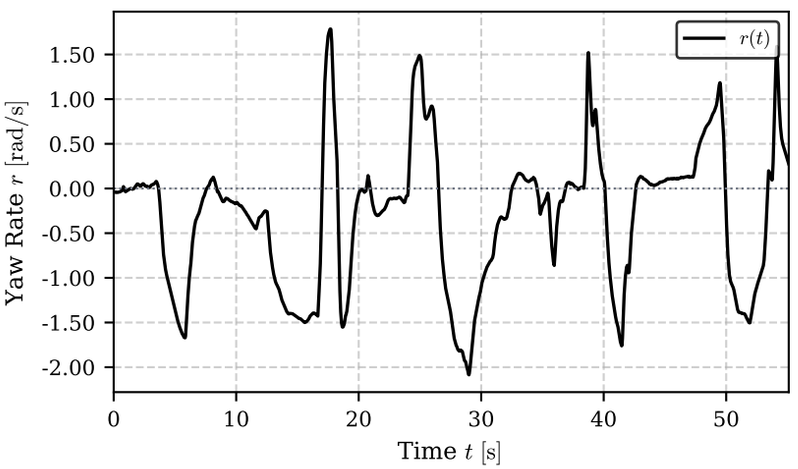

M2 (FL)
Neutral (1500 μs)
M2 (FL) PWM [μs]
Command $[-1, 1]$
Time t [s]

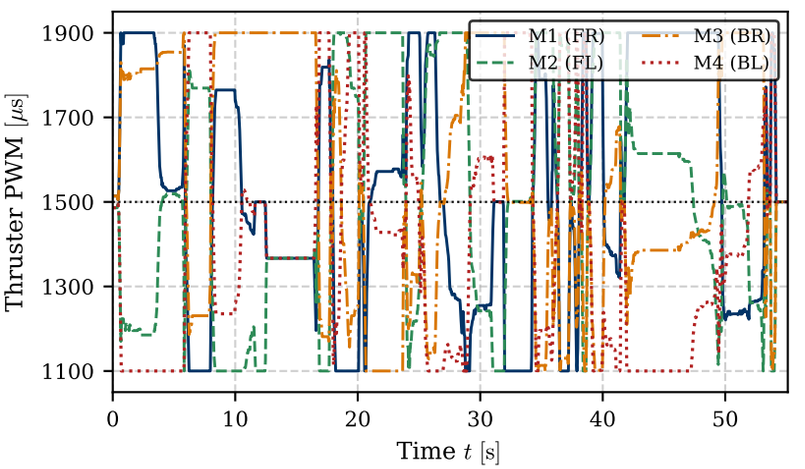

M3 (BR)
Neutral (1500 μs)
M3 (BR) PWM [μs]
Command $[-1, 1]$
Time t [s]

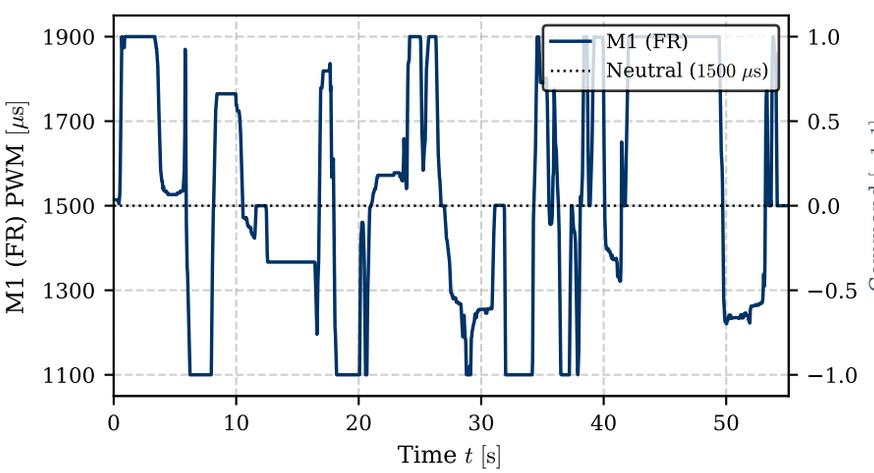

M4 (BL)
Neutral (1500 μs)
M4 (BL) PWM [μs]
Command $[-1, 1]$
Time t [s]

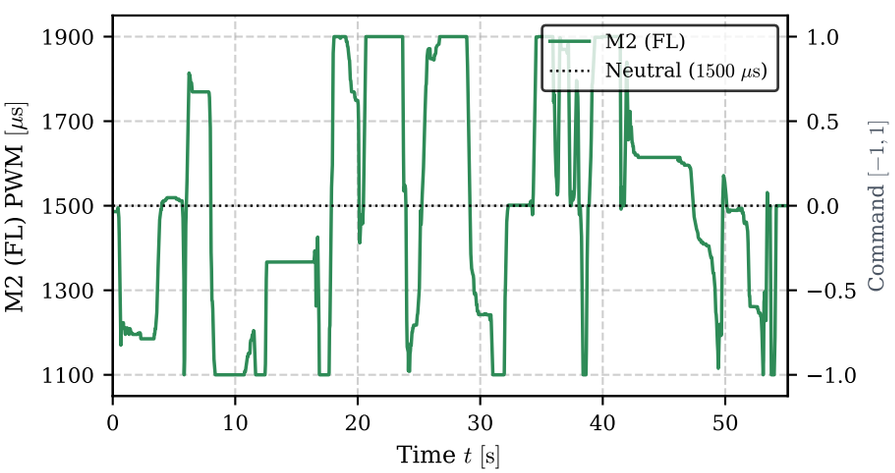

$u(t)$
Surge Velocity u [m/s]
Time t [s]

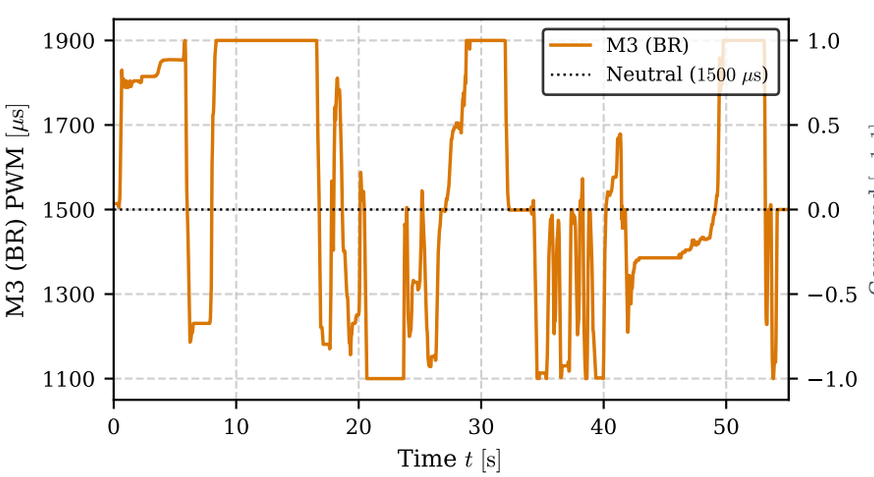

$v(t)$
Sway Velocity v [m/s]
Time t [s]

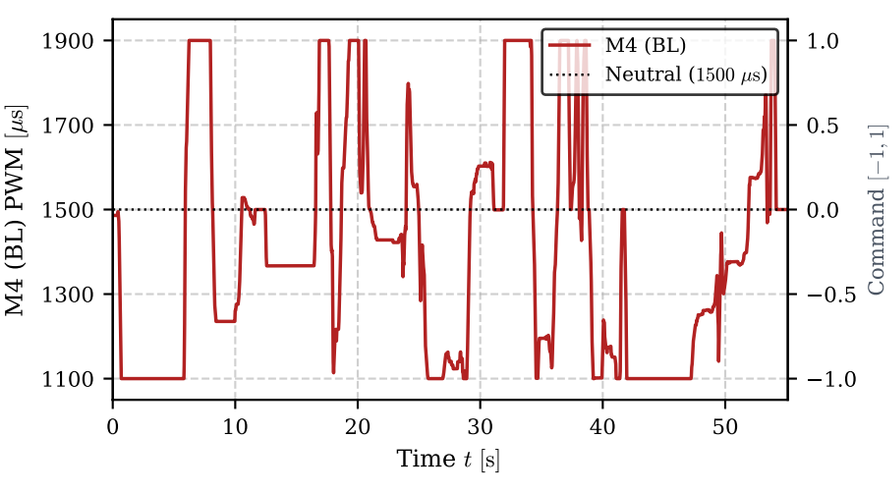

$r(t)$
Yaw Rate r [rad/s]
Time t [s]

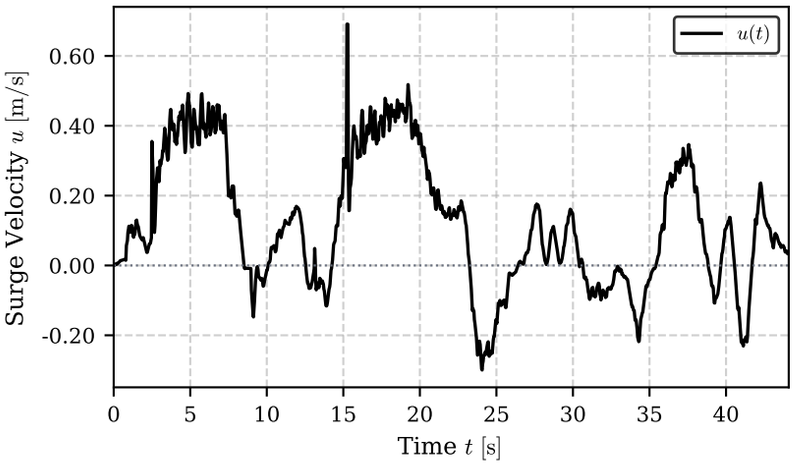

M1 (FR)
M2 (FL)
M3 (BR)
M4 (BL)
Thruster PWM [μs]
Time t [s]

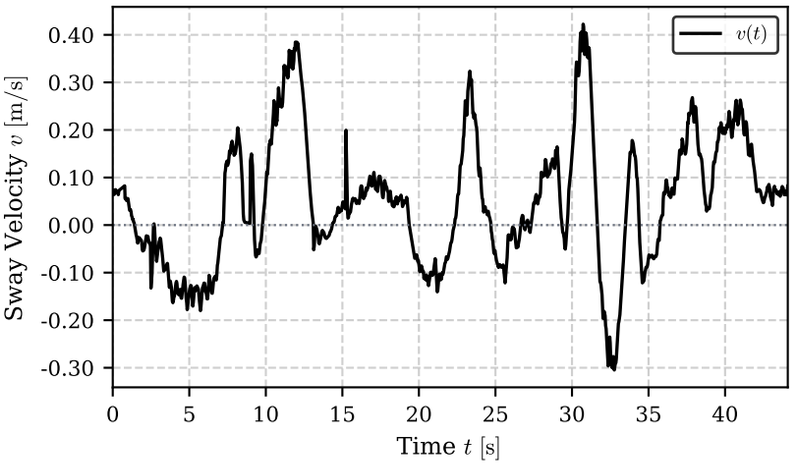

M1 (FR)
Neutral (1500 μs)
M1 (FR) PWM [μs]
Command $[-1, 1]$
Time t [s]

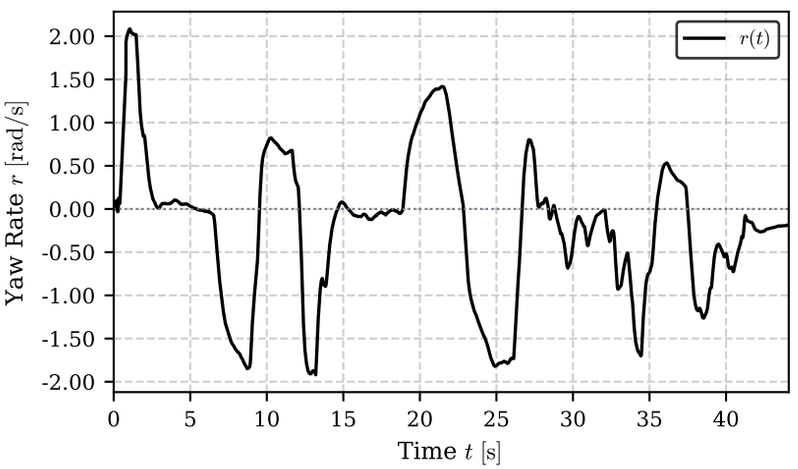

M2 (FL)
Neutral (1500 μs)
M2 (FL) PWM [μs]
Command $[-1, 1]$
Time t [s]

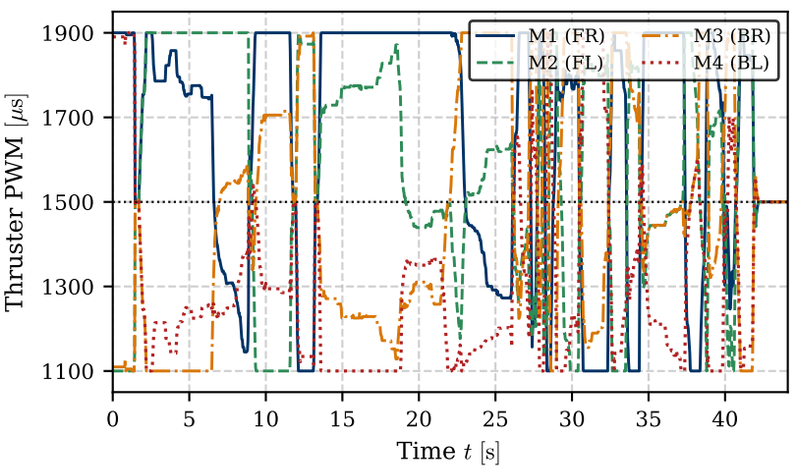

M3 (BR)
Neutral (1500 μs)
M3 (BR) PWM [μs]
Command $[-1, 1]$
Time t [s]

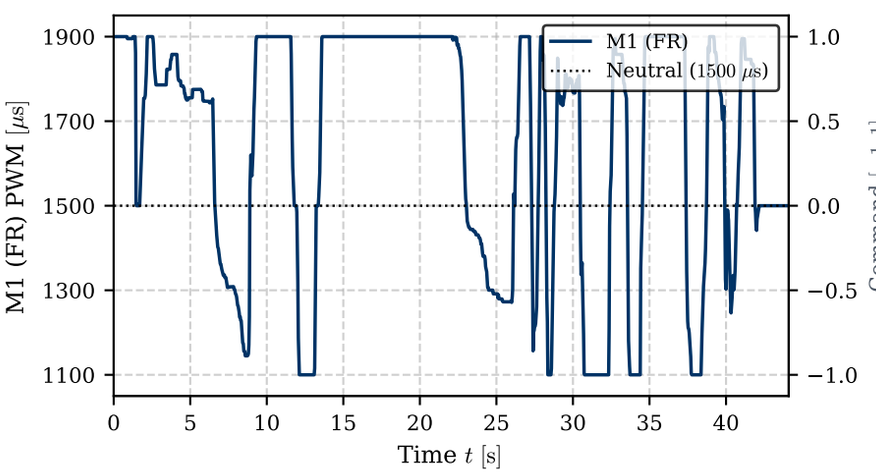

M4 (BL)
Neutral (1500 μs)
M4 (BL) PWM [μs]
Command $[-1, 1]$
Time t [s]

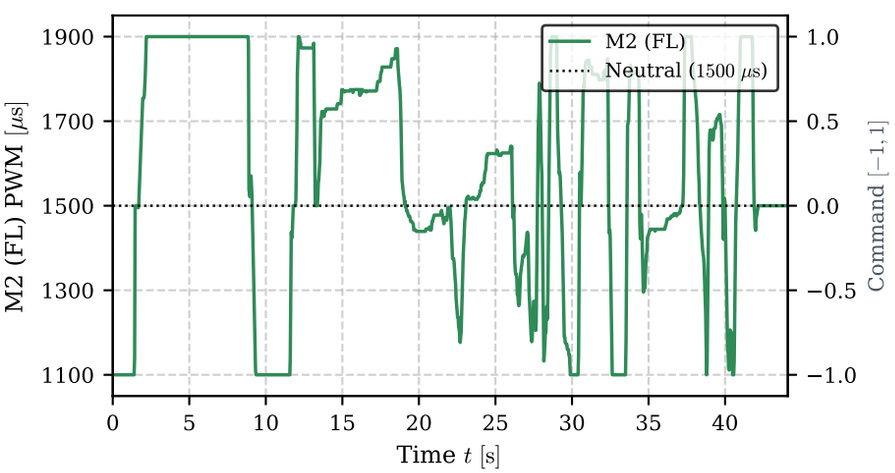

u(t)
Surge Velocity u [m/s]
Time t [s]

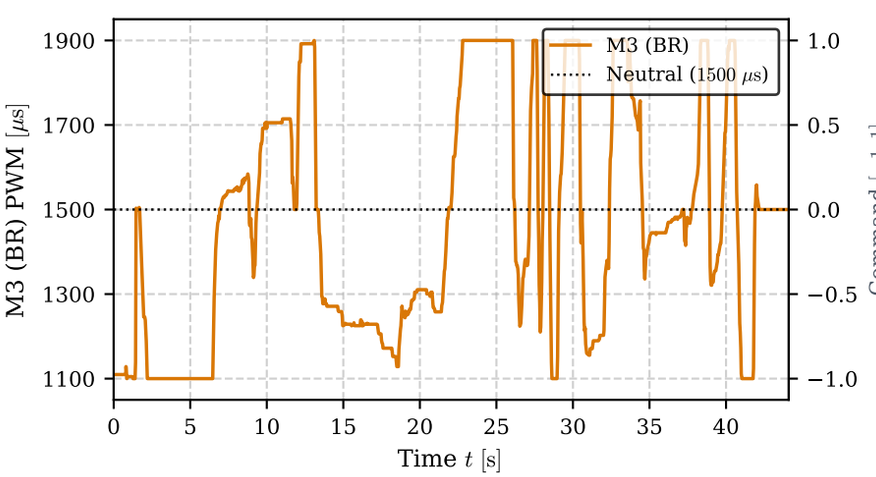

v(t)
Sway Velocity v [m/s]
Time t [s]

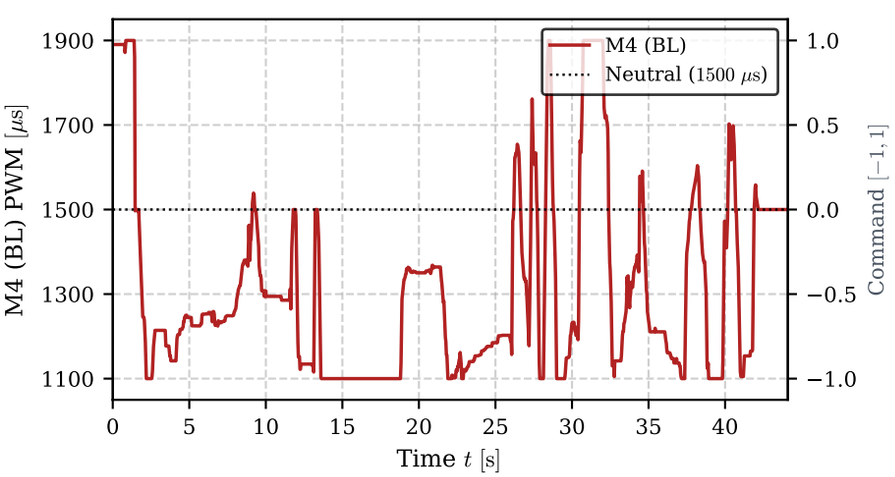

r(t)
Yaw Rate r [rad/s]
Time t [s]

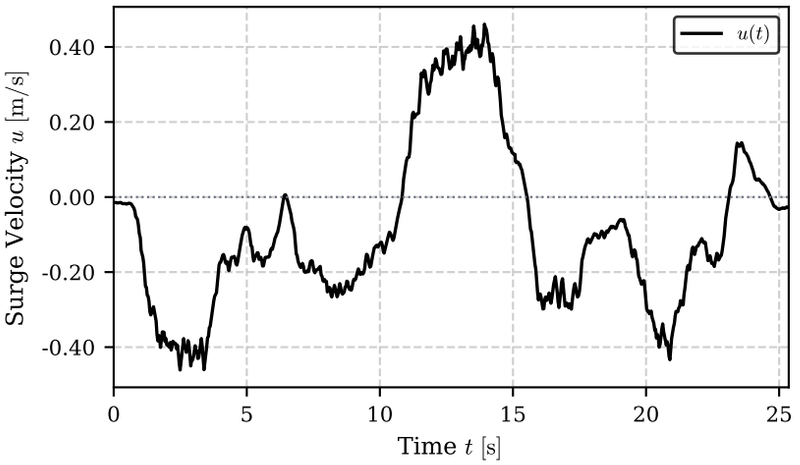

M1 (FR)
M2 (FL)
M3 (BR)
M4 (BL)
Thruster PWM [μs]
Time t [s]

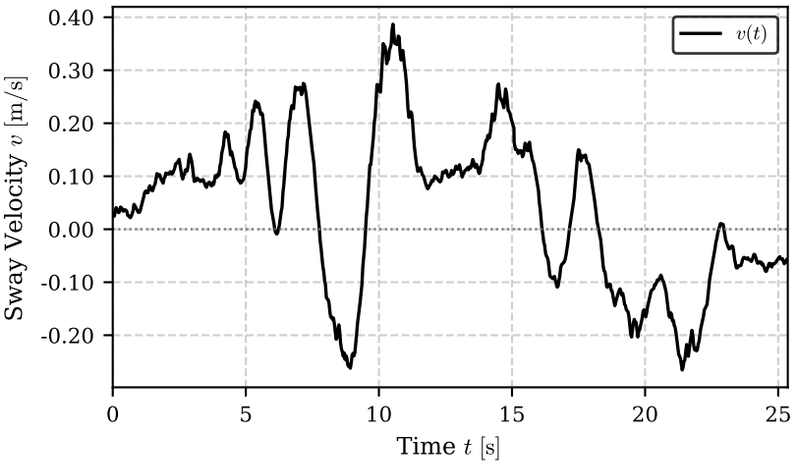

M1 (FR)
Neutral (1500 μs)
M1 (FR) PWM [μs]
Command [−1, 1]
Time t [s]

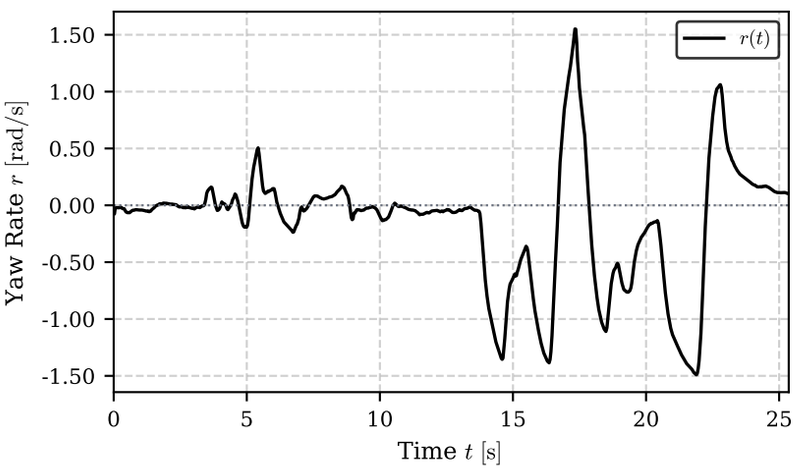

M2 (FL)
Neutral (1500 μs)
M2 (FL) PWM [μs]
Command [−1, 1]
Time t [s]

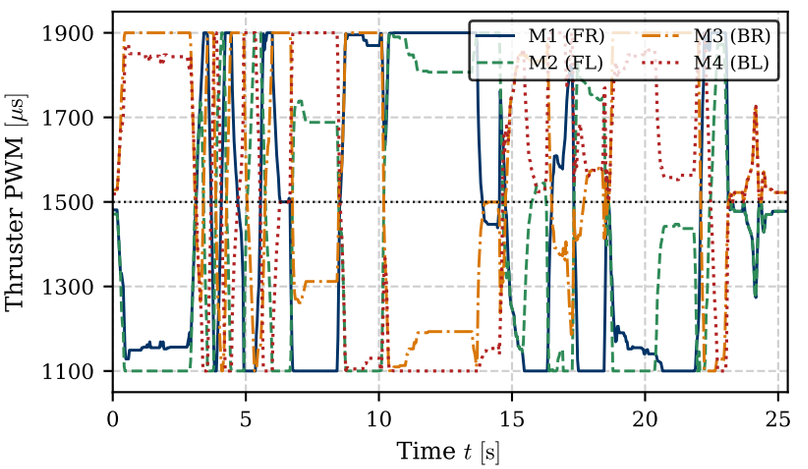

M3 (BR)
Neutral (1500 μs)
M3 (BR) PWM [μs]
Command [−1, 1]
Time t [s]

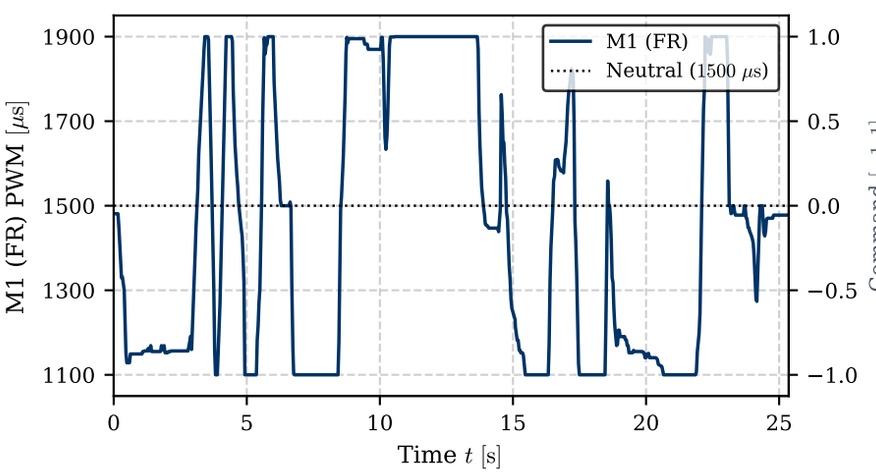

M4 (BL)
Neutral (1500 μs)
M4 (BL) PWM [μs]
Command [−1, 1]
Time t [s]

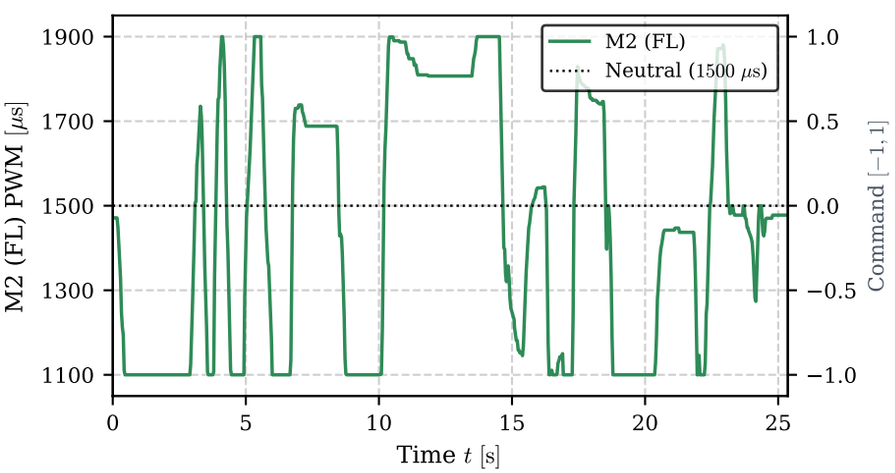

$u(t)$
Surge Velocity u [m/s]
Time t [s]

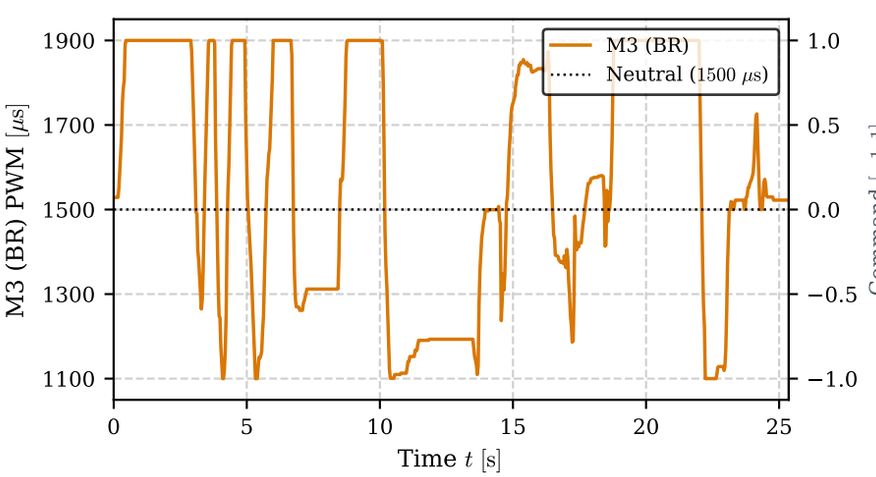

$v(t)$
Sway Velocity v [m/s]
Time t [s]

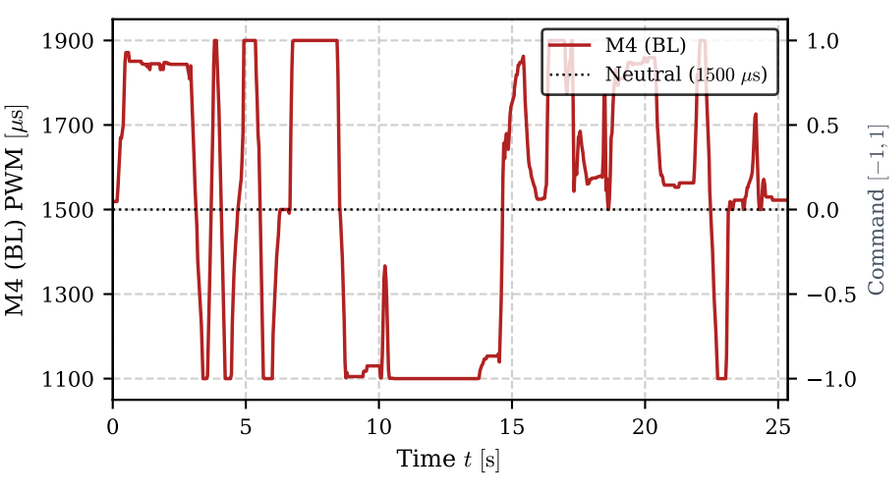

$r(t)$
Yaw Rate r [rad/s]
Time t [s]

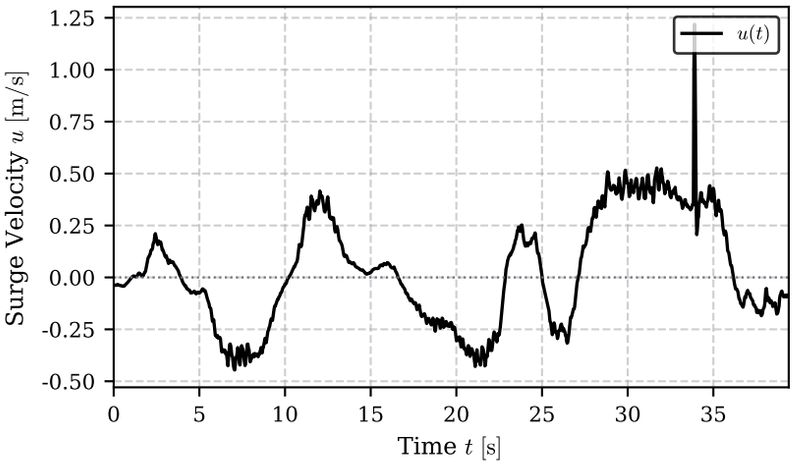

M1 (FR)
M2 (FL)
M3 (BR)
M4 (BL)
Thruster PWM [μs]
Time t [s]

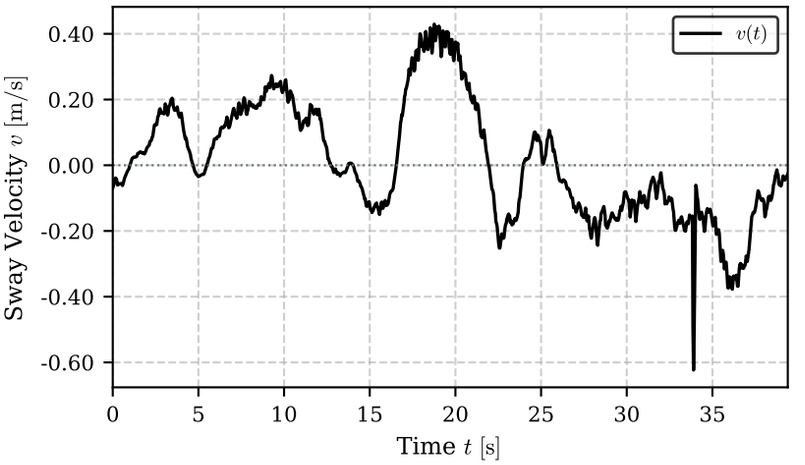

M1 (FR)
Neutral (1500 μs)
M1 (FR) PWM [μs]
Command $[-1, 1]$
Time t [s]

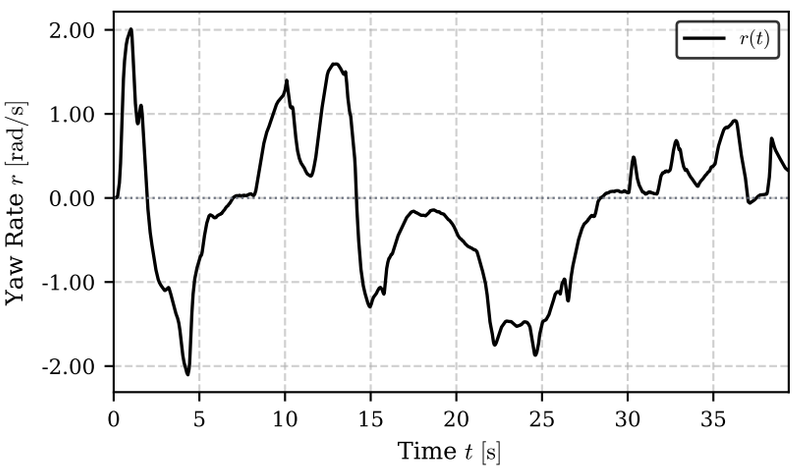

M2 (FL)
Neutral (1500 μs)
M2 (FL) PWM [μs]
Command $[-1, 1]$
Time t [s]

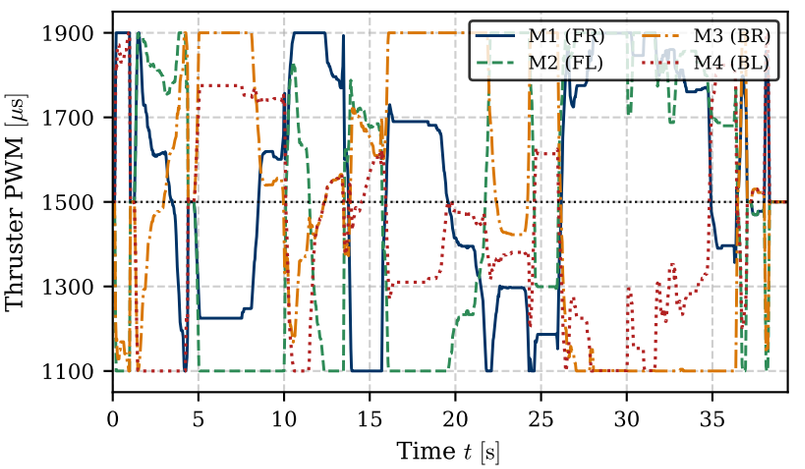

M3 (BR)
Neutral (1500 μs)
M3 (BR) PWM [μs]
Command $[-1, 1]$
Time t [s]

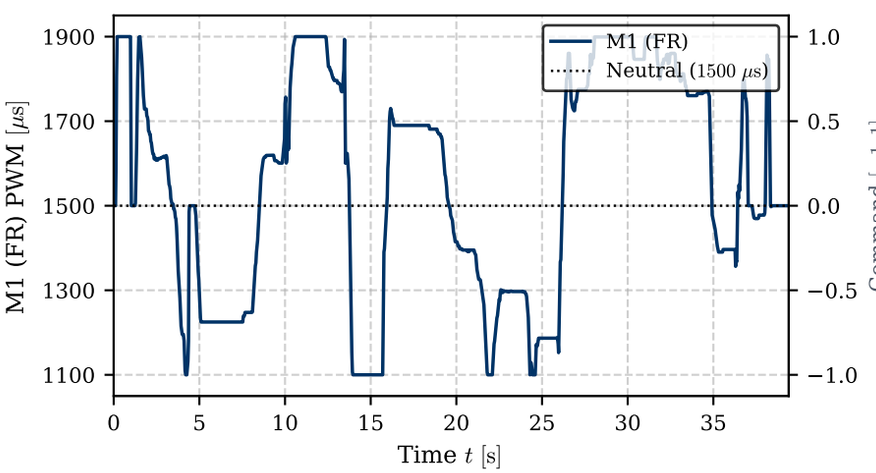

M4 (BL)
Neutral (1500 μs)
M4 (BL) PWM [μs]
Command $[-1, 1]$
Time t [s]

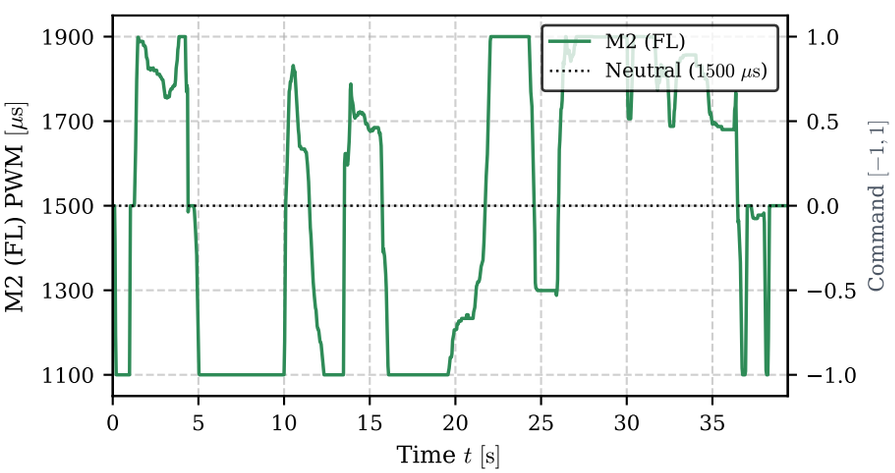

Surge Velocity u [m/s]
$u(t)$
Time t [s]

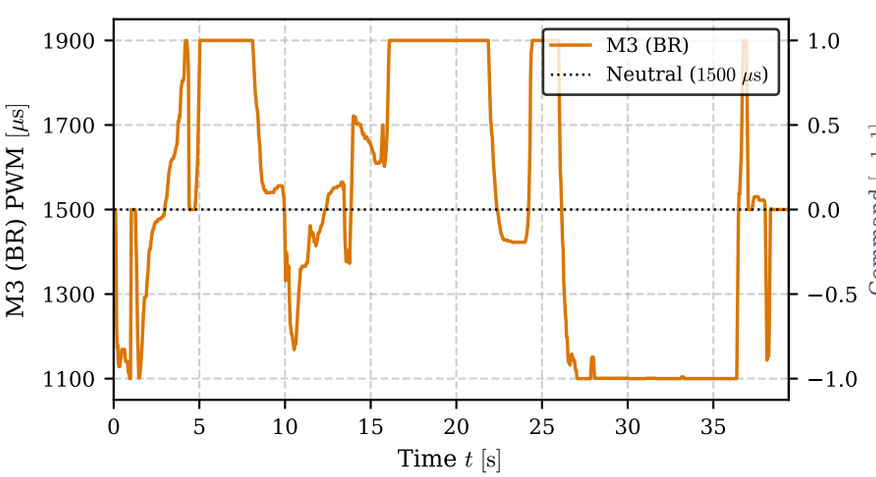

Sway Velocity v [m/s]
$v(t)$
Time t [s]

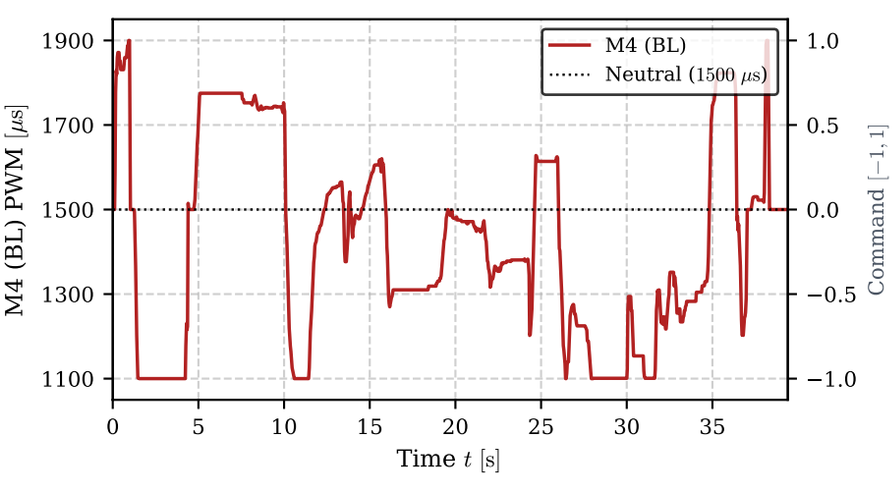

Yaw Rate r [rad/s]
$r(t)$
Time t [s]


Thruster PWM [μs]
M1 (FR)
M2 (FL)
M3 (BR)
M4 (BL)
Time t [s]


M1 (FR) PWM [μs]
Command $[-1, 1]$
M1 (FR)
Neutral (1500 μs)
Time t [s]


M2 (FL) PWM [μs]
Command $[-1, 1]$
M2 (FL)
Neutral (1500 μs)
Time t [s]


M3 (BR) PWM [μs]
Command $[-1, 1]$
M3 (BR)
Neutral (1500 μs)
Time t [s]


M4 (BL) PWM [μs]
Command $[-1, 1]$
M4 (BL)
Neutral (1500 μs)
Time t [s]